Response to the reviewers, second round

Manuscript CAOR-D-26-01885R1, "Optimal-stopping recourse for vehicle routing with simultaneous pickup, delivery, and stochastic demands"

Quang-Vinh Dang, Hoang-Viet Vu, Phuc-Son Nguyen

Dear Editor, dear Reviewers,

We thank the reviewers for their second reading. Reviewer 1 considers the main concerns resolved. Reviewer 2 asks us to align the interpretation of BATON-CF and of the fresh-start bias evidence, and the statistical and implementation descriptions, with what the evidence shows, and points to two features of the code; Reviewer 3 asks four questions about the tables and the cost model. We have checked every implementation point against the code, corrected two conventions in the evaluation and re-run the affected experiments, and revised the text where the reviewers found our claims stronger than the evidence. Section, table and proposition numbers refer to the revised manuscript; a version with the changes marked against revision 1 is provided separately.

Summary of the changes

1. **Baton-cf as an approximation** (Section 3.5). The text now distinguishes the cumulative increment S_k from the physical load X_k, explains what information is retained after unloading, describes BATON-CF as an approximation that reduces rather than removes the dependence bias, gives a sufficient condition for the monotonicity of $F_k(\cdot)$ and checks it empirically.
2. **Fresh-start bias** (Section 3.5, Table A.3). The three estimands are defined, the two measured differences carry instance-level bootstrap intervals, the claim that two biases "nearly cancel" is withdrawn, and the two reference programs are labelled separately.
3. **Two corrections in the evaluation code**, with the affected experiments re-run: routes that depart fully loaded now keep the physical slack $B = 0$, and the physical post-return state is capped at an empty vehicle. Neither changes a conclusion.
4. **Scope and statistics**: the day-type experiment is described as concerning known day types under exchangeable days, its intervals resample instances, ties are reported as the absence of a statistically detectable difference, the over-triggering bound is stated as a sufficient condition, and the exact dynamic program carries a grid-refinement check.
5. **Reviewer 3's questions**: the meaning of "bound" and "reference point", the contrast between the SAA and WDRO gates, relative versus absolute costs in the sensitivity analysis (with absolute costs added to Table 11), and the dispatch fee F_{ho}.
6. **Reproducibility**: the repository state of each submitted version is tagged (`baton-r1`, `baton-r2`).

Reviewer 1

I have examined the revised version and consider that the authors have satisfactorily addressed the main concerns raised in my initial review. [...] Overall, I consider that the revised manuscript represents a substantial improvement over the original submission and that my principal concerns have been adequately resolved.

Response. We thank the reviewer for the careful second reading and for the encouraging assessment. The changes made in this round, in response to Reviewers 2 and 3, narrow several claims further and correct two conventions in the evaluation code; none of them changes a conclusion that the reviewer's comments concerned.

Reviewer 2

1. The remaining information-state approximation in Baton-cf. *[...] Let S_k denote cumulative observed increments without resets and X_k the physical load after returns; following a return at r, $X_k = S_k - S_r$. Under factor dependence, identical (k, X_k) can conceal different S_r and different predictions of remaining demand. [...] The public implementation previously inspected trains conditional models on unreset cumulative sums but supplies reset W online; if this also applies to the submitted version,* Baton-cf *should remain explicitly described as an approximation [...]. The monotonicity of F_k also needs a condition or separate justification [...]. Constancy under independence is a population property, not a guarantee for finite-sample fits or exact policy agreement with* Baton.

Response. The reviewer's reading of the code is correct, and it applies to the submitted version: the continuation and fresh-start models of Baton-cf are fitted on paths indexed by the unreset cumulative increment, and after a return they are evaluated at the reset (physical) load. We have rewritten the description in Section 3.5 along the lines the reviewer suggests. The text now introduces S_k and X_k explicitly and notes that unloading destroys no information: the operator still observes S_k. It explains that under independence the future depends on X_k alone, whereas under factor dependence the factor posterior depends on S_k, so equal (k, X_k) can hide different predictions. Baton-cf conditions the price of a *first* return on its pre-return state (S_k), but applies its continuation models after a return to X_k, as Baton does; it is therefore described as an approximation that reduces bias (iii) for the first return without conditioning later decisions on the full history, and an exact treatment is identified as one on the two-dimensional state (k, X_k, S_k). All statements that Baton-cf "removes" the dependence bias, or removes it "at its source", have been replaced by "reduces".

On monotonicity, the reviewer is right that Proposition 4 does not cover the conditional estimator. The text now gives a sufficient condition (the posterior law of the remaining increments stochastically increasing in S_k, and the cost of the remainder of the day under the downstream policy non-decreasing in those increments, path by path), notes that the second part need not hold for a fitted policy, and describes the isotonic form of $\widehat{F}_k$ as a modelling choice. We then test it with the binned shape test used for C_k, applied to the realised post-return cost under Baton-cf's fitted downstream policy on 5×10^4 paths per route: significant decreases occur in 2.2% of adjacent-bin pairs under independence, where $F_k(\cdot)$ is constant in population and the rate is the false-positive level of the test, and in 0.2% and 0.2% of pairs at $\rho = 0.6$ and 0.9. Finally, the text now states that constancy under independence is a population property: in finite samples the two fits differ, and Baton-cf and Baton need not take the same decisions.

Changes. Section 3.5, paragraphs on the fresh-start value and on deployment selection, and after Eq. (14); new paragraph on the monotonicity of the conditional fresh-start value in Section 4.5; wording in Section 4.1 and Section 4.5.

2. What Table A.3 establishes about fresh-start bias. [...] the claim that the first two biases "nearly cancel" is not established. For the same fitted downstream policy, the in-sample and out-of-sample errors are +0.50% and +0.09% of the handoff price, respectively; their +0.41% difference does not identify the proposed downward in-sample effect. I suggest defining the three estimands and reporting uncertainty for paired differences, or limiting the conclusion [...]. The population cost of a fixed feasible downstream policy bounds the optimum, whereas a finite-sample estimate need not; the high-data program should also be identified as an approximate reference. Finally, reconcile the references with DP^3_{50k} in Section 3.6: the public code previously inspected uses constant-F `fit_dp_actions` for the bias and replacement-F columns, but `fit_dp_actions_cf` for the final performance column.

Response. We agree on every point. The difference of +0.41% indeed has the opposite sign to the in-sample optimism we had invoked, so "nearly cancel" was not supported, and we have withdrawn it. Following the reviewer's suggestion, Section 3.5 now defines three quantities: the optimal F_k; the population value F_k^{pop} of the reset state under the fitted downstream policy; and the in-sample estimate $\widehat{F}_k$. The text states that $F_k^{\text{pop}} \geq F_k$ is a population statement that a finite-sample estimate need not respect, and that the sign of the in-sample error $\widehat{F}_k - F_k^{\text{pop}}$ is not determined a priori. Table A.3 now reports the two measurable differences with 95% bootstrap intervals that resample instances: the in-sample effect $\widehat{F}_k - F_k^{\text{pop}}$ is +0.40% of the handoff price on Salhi–Nagy ([+0.2, +0.6]) and -0.01% on Dethloff ([-0.05, +0.04]); the difference $F_k^{\text{pop}} - F_k^{\text{ref}}$ is +0.09% ([-0.5, +0.6]) and +0.01% ([+0.00, +0.02]). Because F_k^{ref} is itself an estimate, the second difference combines downstream suboptimality with reference error and identifies neither, which the text now says. On the city routes this difference is negative, -1.09% ([-1.7, -0.5]): the fitted policy does better than the reference there, which confirms that the reference is only approximate; the text reports this as well. The conclusion we draw is limited, as the reviewer proposes, to the small change in policy savings when $\widehat{F}_k$ is replaced: on Salhi–Nagy the saving moves from 52.6% to 52.6% with F_k^{pop} and to 52.2% with F_k^{ref} (-0.3 points, [-0.77, -0.02]).

The reviewer's reading of the code is again correct. The bias and replacement columns use the *constant-F* high-data program (`fit_dp_actions`), while the last column, like the main tables, uses the state-conditional program (`fit_dp_actions_cf`). The two are now labelled separately: the reference for the bias columns is called F_k^{ref} and is described as the fresh-start value of $\text{DP}^{3,c}_{50\text{k}}$, the unconditional counterpart of $\text{DP}^3_{50\text{k}}$, and both are described as approximate references.

Changes. Section 3.5 (paragraph on the estimate of the fresh-start value); Table A.3 (new first two columns, intervals, caption); paragraph on the fresh-start value in Section 4.5.

3. Scope and uncertainty of the new promotion-day experiment. [...] a fixed ordinary/promotional mixture can remain exchangeable across days. I suggest narrowing the abstract and section claims accordingly; this experiment does not establish robustness to temporal drift or serial dependence [...]. The interval pooling 80 Det/SAA plans from 40 instances should analyse each gate separately or resample by instance with both plans retained together. For the deliver-only comparison, the Table 5 interval $[-6.9, +2.4]$ percentage points supports "no statistically detectable difference," rather than an established performance tie.

Response. We agree. The abstract, the introduction and Section 4.5 now speak of *known day types* rather than non-exchangeable days, and the section states that the days are drawn independently from a fixed mixture, so they remain exchangeable, and that the experiment focuses on known day types, with temporal drift and serial dependence left for future work; the conclusion says the same. The interval now resamples instances with both plans kept together, and we also

report each gate separately: the day-type-specific fits differ from the pooled fit by -0.5 points overall ([-0.7, -0.3]), +0.1 on DET plans ([-0.1, +0.3]) and -1.1 on SAA plans ([-1.5, -0.8]). Finally, wherever an interval contains zero we now write that the difference is not statistically detectable rather than that the policies are tied, including the deliver-only comparison, the deterministic gate and the dependence rows.

Changes. Abstract; introduction; Section 4.5 ("Known day types"); conclusion; caption of Table 9; wording throughout Section 4.

> *4. Physical-reset conventions and evaluation code. [...] I suggest explaining what a known departure load reveals about future deliveries when actual quantities are loaded, or how deliveries exceeding loaded cargo are handled when nominal quantities are used, so that $L_0 - D_{\le k}$ has a consistent physical meaning. The public `run_realistic_eval.py` and `route_setup` in `run_baton_r1.py` previously inspected replace $B = Q - L_0$ with an empirical peak quantile when $B \le 0$, although zero slack is a valid fully loaded departure. State whether this branch is activated in the reported experiments and how many routes it affects. [...] A fixed R1 repository commit or archive would allow verification against the submitted implementation.*

Response. Thank you; both points led to corrections. On the loading convention, our simulator uses *nominal* loading: the vehicle departs with the planned delivery volume $L_0 = \sum_k \bar{d}_k$ and the realised deliveries are revealed on service. Section 3.1 now states this, reports how often the day's deliveries exceed the cargo loaded (on 47% of route-days, by 10% of L_0 on average), and explains that such shortfalls are outside the model: they are not priced and no policy affects them, so they do not change the comparison of policies. It also explains that loading the realised volume would make L_0 random but known at departure and reveal the day's delivery total before the first stop, a different information structure that we do not study. For the physical reset this means that the undelivered cargo is $\max\{L_0 - D_{\le k}, 0\}$; the submitted implementation did not cap it, so the physical post-return state could fall below $-L_0$. The state is now $x_k = -\min\{D_{\le k}, L_0\}$, and the physical-reset column of Table A.3 was re-computed with it (58.2% on Salhi–Nagy and 54.6% on Dethloff). The conclusion that the conservative reset understates the value of the return is unchanged.

On the slack, the branch was activated. No route of any reported experiment departs over capacity, but 16 routes depart exactly full ($B = 0$): one Salhi–Nagy route and fifteen routes of the city instances and their twins (none of the Dethloff routes, none of the routes of the RL comparison, and none of the city SAA plans). For these routes the code replaced $B = 0$ by an empirical peak quantile. We now keep the physical slack $B = 0$ in all evaluation code, so that such a route breaches as soon as its running load rises, and we re-ran every affected experiment: the Salhi–Nagy and city benchmarks of Table 5, the fresh-start, over-triggering, data-budget and standby-pool experiments. The mean savings of the implementable policies moved by at most 0.7 percentage points (on the uniform city twins) and those of the reference points by at most 1.7 points (the three-action clairvoyant on the 50% deliver-only twins); no ranking or conclusion changed, and the manuscript reports the re-run numbers. Last, the repository now carries the tag `baton-r1` for the state that produced revision 1, and `baton-r2` for this revision, so that the code can be checked against each submitted version; the Data availability statement says so.

Changes. Section 3.1 (loading convention, fully loaded departures); Section 3.5 (physical reset); Table 5, Table A.3, Table 6, Table 10 and Table A.4 re-computed; Data availability.

> *(a) Proposition 3 and Table 6 address over-triggering. I suggest replacing "good enough exactly when" with a sufficient-condition statement: a small upper bound guarantees a small*

loss but is not necessary. Interpret the numerical verification with the qualification that the policy labelled optimal is estimated from 50,000 paths.

Response. Done. The text after Proposition 3 now presents the bound as a sufficient condition for the myopic threshold to be good enough, and says that it is not necessary because the bound need not be tight. The discussion of Table 6 and its caption now state that the rule labelled optimal is itself an estimate from 5×10^4 paths, so that the table verifies the proposition up to that estimation error.

Changes. Paragraph after Proposition 3; Section 4.4; caption of Table 6.

(b) Table 8 is welcome. Since exact-optimum percentages now appear in the abstract, I suggest adding a small grid-refinement or truncation check for the optimal values. Agreement between simulated and grid-based reactive costs alone does not establish the error in the optimal policy value.

Response. We agree, and we have added the check. For all 627 routes of Table 8 we solved the handoff-only and the three-action programs on grids of 1,500, 3,000 and 6,000 load cells, and with the lower truncation and the support of the demand distributions widened (from 6 to 10 standard deviations and from 12 to 20 coefficients of variation). Values of single routes move by up to 2.91% between 1,500 and 6,000 cells (median 0.4%; 1.20% between 3,000 and 6,000 cells), but the discretisation affects the reactive, handoff-only and three-action values alike, so the pooled exact savings of Table 8, and with them the percentages in the abstract, change by at most 0.01 percentage points across all grids and truncations. The text and the caption of Table 8 now report this check.

Changes. Paragraph on the exact benchmark in Section 4.5; caption of Table 8.

(c) I suggest stating Algorithm 1's action-specific updates explicitly, distinguishing realised continuation-path costs from the estimated $R_k + F_k$ inserted upon return. Also acknowledge that the equal-data three-action DP splits its training sample while Baton reuses it, so the comparisons reflect sample allocation as well as estimator differences.

Response. Done. Algorithm 1 now spells out the three updates: a path whose estimated continuation cost exceeds the handoff price is assigned that price; a path for which the return is cheapest is assigned the estimated $R_k + F_k$; and a path that continues keeps its realised cost along the path. The protocol paragraph in Section 4.1 now notes that DP^3_N splits its 10^3 training days between its two estimation steps whereas Baton uses all of them for both, so that the comparison reflects the allocation of the sample as well as the estimators.

Changes. Algorithm 1; "Protocol" paragraph in Section 4.1.

Reviewer 3

1. In Table 1, both oracle-ho and oracle3 are labeled as 'bounds', however, in subsequent simulation studies, as shown in Table 2, Table 11, Fig. 5, and Fig. 8, only oracle3 appears as a bound, oracle-ho becomes a reference point, is the reference point equivalent to bound?

Response. Thank you for pointing out the inconsistency. The two words are not equivalent, and we now use them consistently. A clairvoyant policy is a *bound* for a class of policies when

no policy in that class can do better on any day. The handoff-only clairvoyant (oracle-ho) knows the day's demands but may only hand off; it is therefore a bound for the handoff-only policies (the thresholds, the rule-based triggers, the rollouts, BATON-HO), but not for policies that may also return to the depot, which can prevent a breach more cheaply than any handoff and can therefore exceed it, as BATON does on four of the six gates. For those policies oracle-ho is only a *reference point*. The three-action clairvoyant (oracle^3) is a bound for every policy in the paper. Table 1 now says for which policies each clairvoyant is a bound, Section 3.6 defines the two terms, and the captions of the results tables use them in this sense; the reference-point columns of the tables group the clairvoyant policies with the high-data programs, which are neither bounds nor competitors.

Changes. Table 1; Section 3.6; caption of Table 2.

2. In Sec 4.1 planning regimes, it claims that 'the WDRO gate is the SAA gate at a tighter threshold', in Table 2, it can be seen that in SAA gate, the value of BATON (53.7) is much higher than that of oracle-ho (43.5), while in WDRO gate, the value of BATON (41.2) is almost the same as that of oracle-ho (42.0). Meanwhile, the value of BATON in SAA gate (53.7) is much higher than that of WDRO gate (41.2). What are the explanation for these?

Response. The values in Table 2 are savings in per cent of each gate's own reactive cost, and that cost differs by a factor of about five between the two gates. The tighter threshold of the WDRO gate produces much safer plans: the reactive policy costs 2.9 per plan-day on WDRO plans against 16.5 on SAA plans, and at least one route of a plan breaches on 2.3% of days against 10.0%. In absolute terms BATON is far cheaper on WDRO plans (1.9 against 8.3 per plan-day), but there are fewer breaches to prevent, and a thousand training days contain few of them; this is the rare-breach regime in which BATON falls furthest short of what more data would allow (41.2% against 47.8% for the high-data three-action program, compared with 53.7% against 56.4% on SAA). The handoff-only clairvoyant knows the day's demands and saves a similar share on both gates (42.0% and 43.5%), so BATON exceeds it by a wide margin on SAA plans but not on WDRO plans. We have added this explanation to Section 4.2.

Changes. New paragraph in Section 4.2.

3. In Table 11, it shows that the value of BATON with a cheap standby (Fsb=10, BATON=52.7) is much higher than that with a dear standby (Fsb=35, BATON=42.6), while the value of BATON with a cheap emergencies (Femg=25, BATON=40.1) is much low than that with a dear emergencies (Femg=60, BATON=55.2). Does this mean that cheap standby is better, and cheap emergencies are not better?

Response. No: in absolute terms both are better. The values in Table 11 are savings relative to the reactive cost of the same configuration, and the reactive cost itself depends on the emergency price. BATON's expected daily cost per plan is 52.7 at the default prices, 45.8 with cheap standby and 48.3 with cheap emergencies, against 60.9 with dear standby and 57.7 with dear emergencies. The savings move in opposite directions for a different reason. Cheap standby makes the interventions cheaper while the reactive cost stays at 78.4, so the saving rises. Cheap emergencies lower the reactive cost itself, to 64.2, so there is less to save, and dear emergencies raise it, to 97.3, so intervention recovers a larger share of a larger bill. We have added the reactive and BATON daily costs as two columns of Table 11 and this explanation to Section 4.8.

Changes. Table 11 (two new columns, caption); Section 4.8.

4. In Eq. (4), does the Fho represent dispatch fee?

Response. Yes. F_{ho} is the dispatch fee, the fixed charge for calling out a standby vehicle. The symbol was not named in the text, which now says so where Eq. (4) is introduced.

Changes. Section 3.1.

We thank the reviewers again for their time and for comments that have made the paper more precise.

Sincerely,
Quang-Vinh Dang (corresponding author), on behalf of all authors